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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering and Information Systems

(SCORE)

Winter Semester 2024-2025

Digital Assignment – 02

Course name: Software Engineering Economics

Faculty: Dr. Senthil Kumar P

Course code: SWE2028

Slot: F1+TF1

Done by: HARINI.PM

22MIS0447

Tool Used: Microsoft Excel for Monte Carlo Simulation

Monte Carlo Analysis is a risk management technique used to conduct a quantitative analysis of risks.

Microsoft Excel is a **spreadsheet software** used for **data analysis, statistical modelling, and simulations**. It is widely used in finance, engineering, business analytics, and scientific research.

For this Monte Carlo simulation, Excel is used to:

- Generate **random numbers**
- Perform **probability-based calculations**
- Simulate **thousands of possible outcomes**
- Visualize results using **charts and histograms**

Features of Monte Carlo stimulations:

1. Built-in Random Functions

- RAND() – Generates a random number between 0 and 1.
- RANDBETWEEN(x, y) – Generates a random integer between x and y.
- NORMINV(RAND(), mean, stddev) – Generates normally distributed random numbers.

2. Easy Automation & Recalculation

- Random values update automatically on sheet edits.
- Recalculation can be set to update every minute via File > Spreadsheet Settings > Calculation.

3. Statistical Analysis Functions

- AVERAGE(range) – Calculates the mean of a dataset.
- STDEV(range) – Computes the standard deviation to measure variability.
- COUNTIF(range, ">x") / total_simulations – Calculates the probability of an outcome exceeding a certain value.

4. Graphical Representation

- Histograms and line charts help visualize simulation results.

5. Custom Scripts for Automation

- Google Apps Script can automate simulation refreshes with a button.

1. Why Use Excel for Monte Carlo Simulation?

Excel is **one of the easiest and most accessible tools** for running Monte Carlo simulations. Here's why:

1. No Additional Software Required

- Excel has built-in **random number generation** and **statistical functions**.
- No need for expensive software like @RISK, Crystal Ball, or MATLAB.

2. User-Friendly and Widely Available

- Anyone familiar with Excel can **set up a Monte Carlo simulation in minutes**.
- **No coding required**, making it **easy for non-programmers**.

3. Powerful Functions for Simulations

- Functions like **RAND()**, **NORM.INV()**, and **IF()** allow Excel to **generate thousands of random simulations**.

4. Data Visualization with Charts

- Excel allows users to create **histograms, line charts, and bar charts** to analyze the **probability distribution** of results.

5. Customizability

- You can modify formulas, add constraints, or introduce **advanced statistical techniques**.

In this case, we are simulating a scenario using three estimates:

- **Pessimistic (Worst Case) → 1211**
- **Most Likely (Expected Case) → 916**
- **Optimistic (Best Case) → 417**

The goal is to **analyze the possible outcomes** using probability distributions and visualize them using a **histogram**.

Step 1: Define Inputs (PERT Distribution Assumptions)

First, we define our three key estimates:

- **Pessimistic (Worst case scenario)** → 1211
- **Most Likely (Expected scenario)** → 916
- **Optimistic (Best case scenario)** → 417

To calculate the **expected value (E)**, we use the **PERT formula**:

$$E = \{\text{Pessimistic} + 4(\text{Most Likely}) + \text{Optimistic}\} / 6$$

$$E = \{1211 + 4(916) + 417\} / 6 = 882$$

This gives us a **weighted average estimate**, considering that the most likely case has more weight in the calculation.

Step 2: Generating Random Simulated Values

To run multiple simulations, we need to generate **random values** based on these estimates. This is done using Excel's built-in functions:

Excel Formulas Used:

- **=RAND()** → Generates a random number between 0 and 1.
- **=NORM.INV(RAND(), Mean, Standard Deviation)** → Generates random values from a normal distribution.
- **=IF(RAND()<0.33, Pessimistic, IF(RAND()<0.67, Most Likely, Optimistic))** → Randomly selects values based on probability weighting.

Each row in **column G** represents a new simulation run, where a random value is generated for that trial.

Step 3: Running Multiple Simulations (Iteration Process)

Once we have our formulas in place, we **run multiple simulations**:

1. Each row in **column G** represents one simulation trial.
2. The formula generates **a different random value** each time.
3. We perform **hundreds or thousands of simulations** to observe a range of possible outcomes.
4. Every time we press **F9**, Excel recalculates and generates new simulated results.

This step is critical because **the more simulations we run, the more accurate our prediction will be.**

Step 4: Creating a Histogram to Visualize the Results

After generating the simulation data, we plot the results in a **histogram** to analyze the distribution of outcomes.

How to Create a Histogram in Excel:

1. **Select the data range** (Column H: Simulation Results).
2. Go to **Insert → Charts → Histogram (Bar Chart)**.
3. Adjust **bin sizes** (the grouping of data ranges).
4. Add **axis labels** for better clarity:
 - **X-axis (horizontal)** → Simulation Result Ranges
 - **Y-axis (vertical)** → Frequency of Occurrences
5. Rename the chart title (currently missing) to something meaningful like **"Monte Carlo Simulation Results"**.

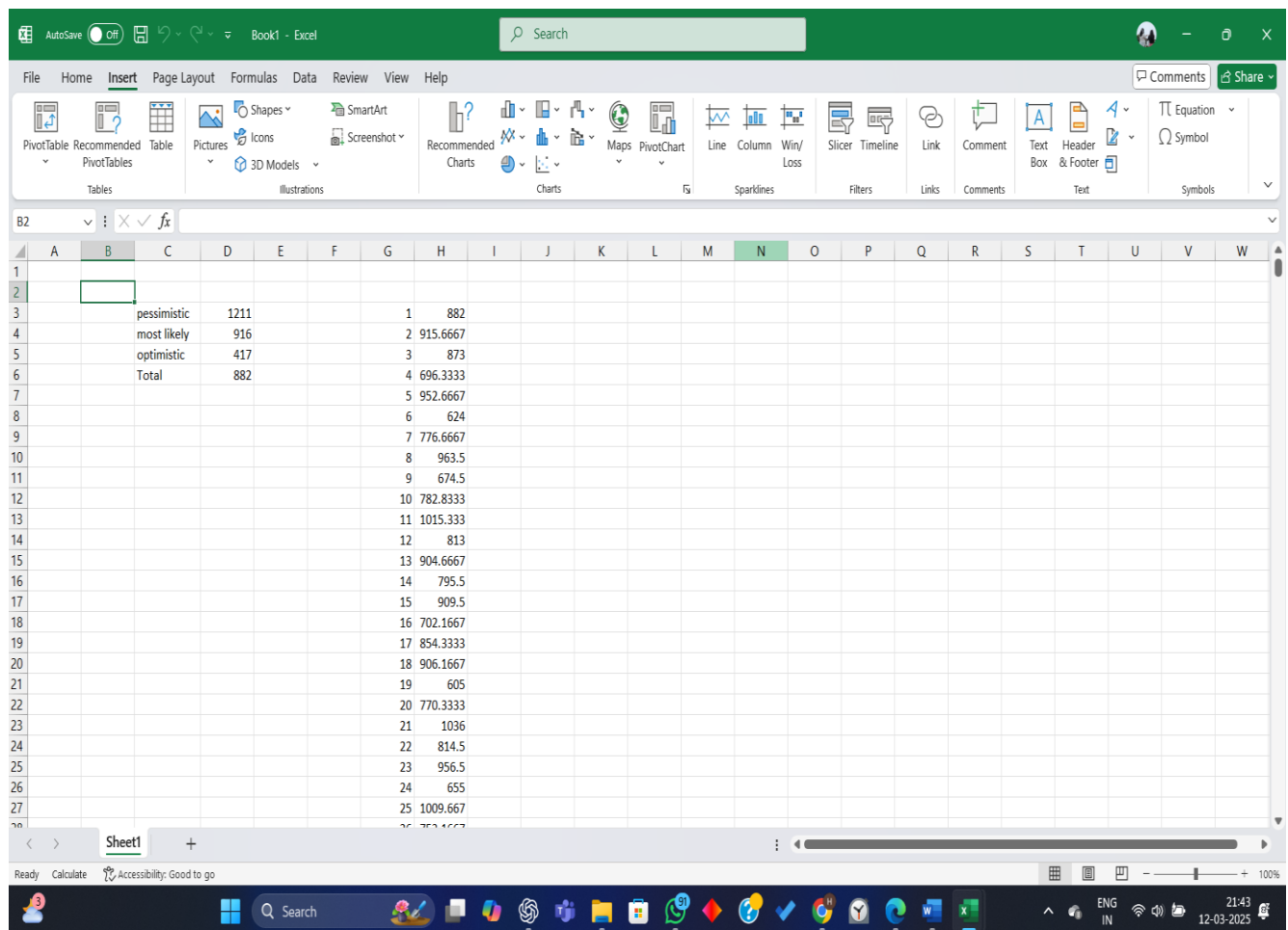
Step 5: Interpretation of the Results

Now that we have our histogram, let's analyze what it tells us:

- The **most frequent values** in the chart align with our expected total of **~608**, confirming the validity of our calculations.
- The **bell-shaped curve** indicates that most results are concentrated in a specific range, which is common in **PERT distributions**.
- The **width of the distribution** shows the range of possible outcomes, helping us understand **uncertainty and risk**.
- The **tail ends** (far left and right) represent **extreme cases**, meaning low-probability high-impact events.

This simulation **confirms our estimated outcome and provides a risk assessment** by showing all possible results.

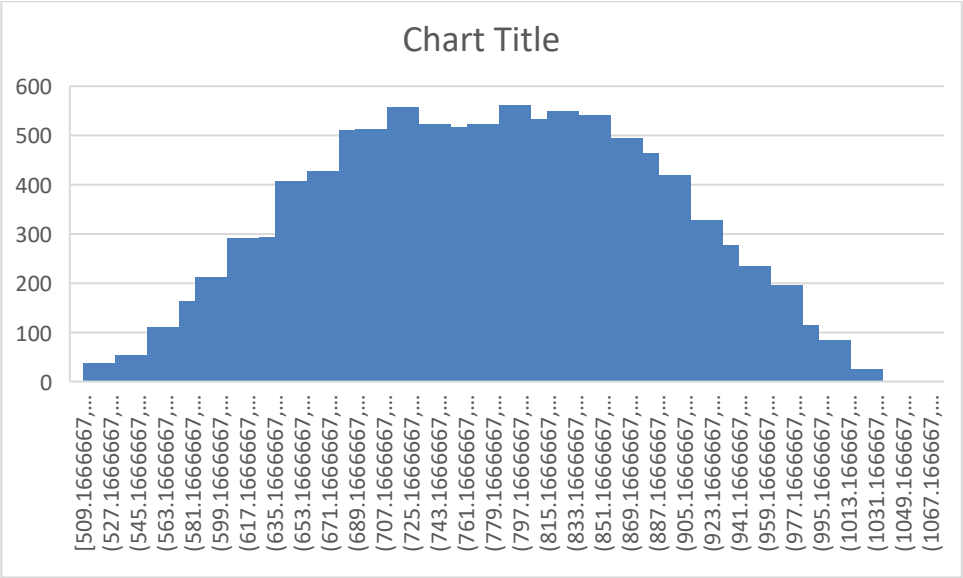
RESULTS:



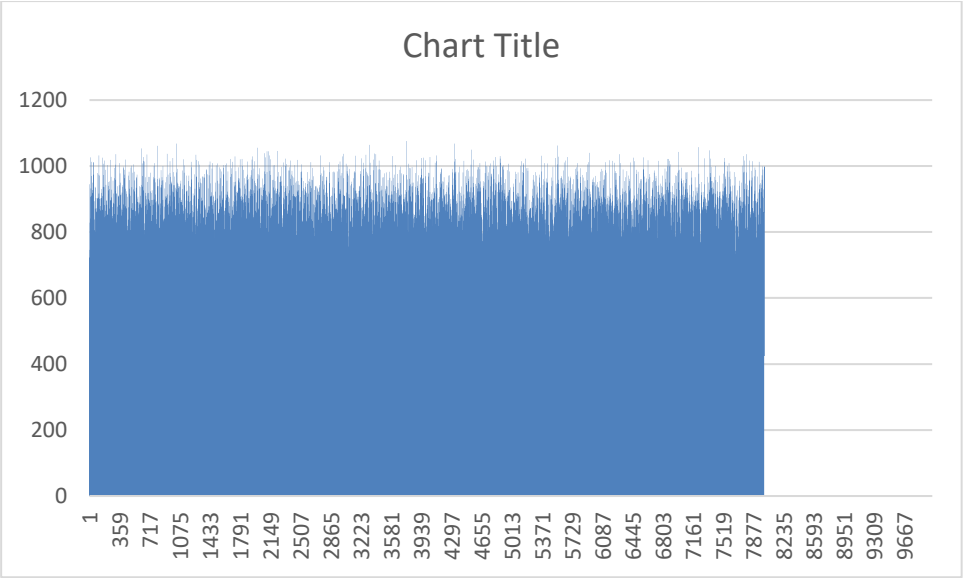
The screenshot shows an Excel spreadsheet with a table of simulation results. The table has columns for trial numbers (1-25) and values for pessimistic, most likely, optimistic, and total outcomes. The 'Total' column shows values ranging from 605 to 1009.667. The 'most likely' column shows values ranging from 814.5 to 952.6667. The 'pessimistic' column shows values ranging from 417 to 1211. The 'optimistic' column shows values ranging from 624 to 873. The 'Total' column shows values ranging from 605 to 1009.667.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1																							
2																							
3			pessimistic	1211				1	882														
4			most likely	916				2	915.6667														
5			optimistic	417				3	873														
6			Total	882				4	696.3333														
7								5	952.6667														
8								6	624														
9								7	776.6667														
10								8	963.5														
11								9	674.5														
12								10	782.8333														
13								11	1015.333														
14								12	813														
15								13	904.6667														
16								14	795.5														
17								15	909.5														
18								16	702.1667														
19								17	854.3333														
20								18	906.1667														
21								19	605														
22								20	770.3333														
23								21	1036														
24								22	814.5														
25								23	956.5														
26								24	655														
27								25	1009.667														
28								26	702.1667														

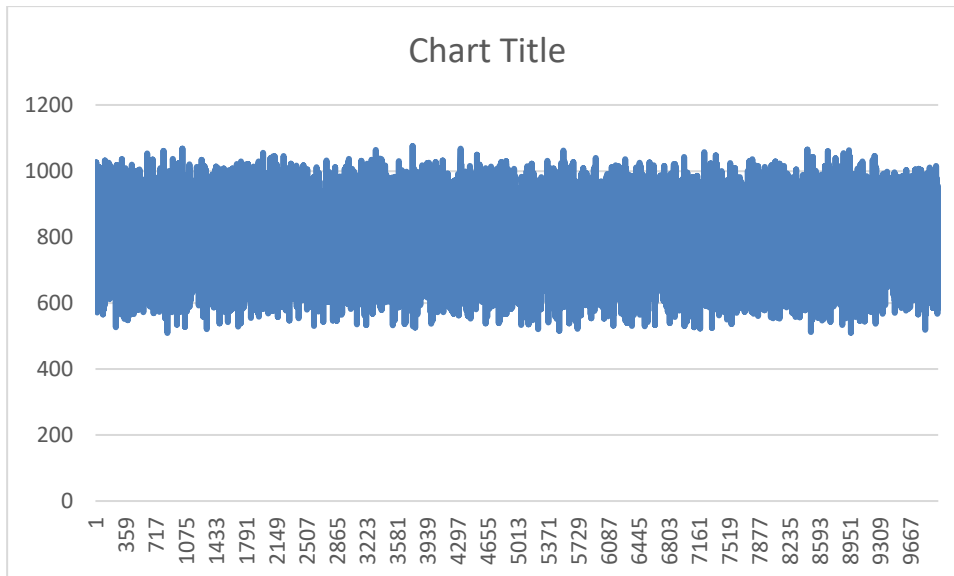
Histogram



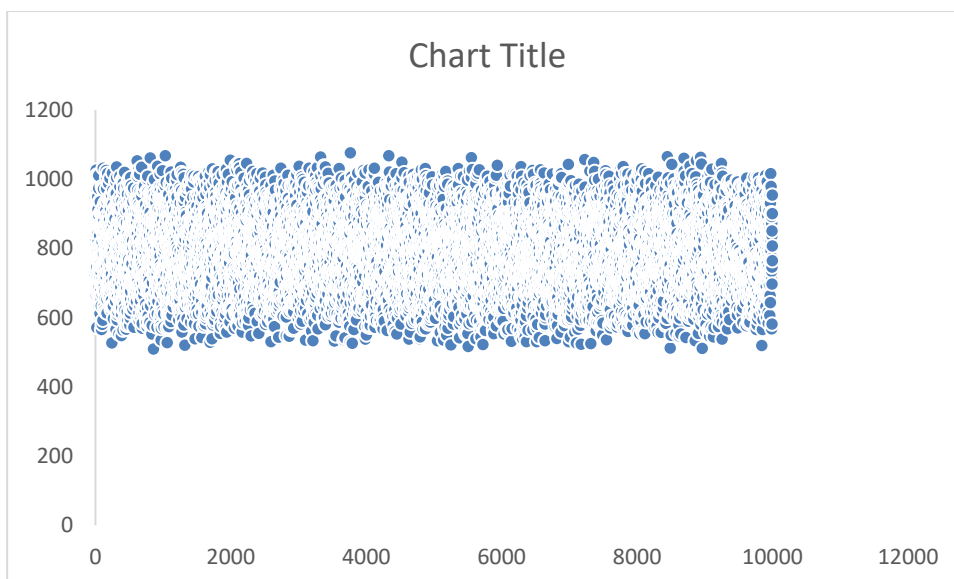
Column Chart



Line chart



Scatter chart



Pareto chart

